

习题九

1.

~~B C D~~

~~K Z₈ Z₄ Z₂ Z₁~~

BCD 码

	K	S ₈	S ₄	S ₂	S ₁
0.	0	0	0	0	0
1.	0	0	0	0	1
2.	0	0	0	1	0
	0	0	0	1	1
	0	0	1	0	0
	0	0	1	0	1
	0	0	1	1	0
	0	0	1	1	1
	0	1	0	0	0
	0	1	0	0	1
	1	0	0	0	0
	1	0	0	0	1
	1	0	0	1	0
	1	0	0	1	1
	1	0	1	0	0
	1	0	1	0	1
	1	0	1	1	0
	1	0	1	1	1
	1	1	0	0	0
	1	1	0	0	1

补码

	R ₈	R ₄	R ₂	R ₁
	0	0	0	0
	1	1	1	1
	1	1	1	0
	1	1	1	1
	1	1	1	0
	1	1	0	1
	1	1	0	0
	1	1	0	1
	1	0	1	1
	0	0	0	0
	0	1	1	1
	0	1	1	0
	0	1	1	0
	0	1	1	0
	0	1	0	1
	0	1	0	0
	0	1	0	0
	0	0	1	1

$$R_8 = K'(S_8 + S_4 + S_2 + S_1) + K S_8' S_4' S_2' S_1'$$

$$= K \oplus (S_8 + S_4 + S_2 + S_1)$$

$$R_8 = [S_4' S_2' (S_8' S_1' + S_8 S_1)]'$$

$$= S_4 + S_2 + (S_8 \oplus S_1)$$

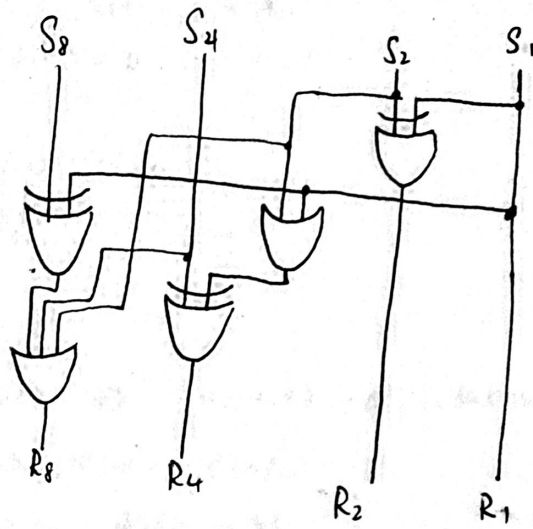
$$R_4 = S_4' (S_2 + S_1) + S_4 S_2' S_1'$$

$$= S_4 \oplus (S_2 + S_1)$$

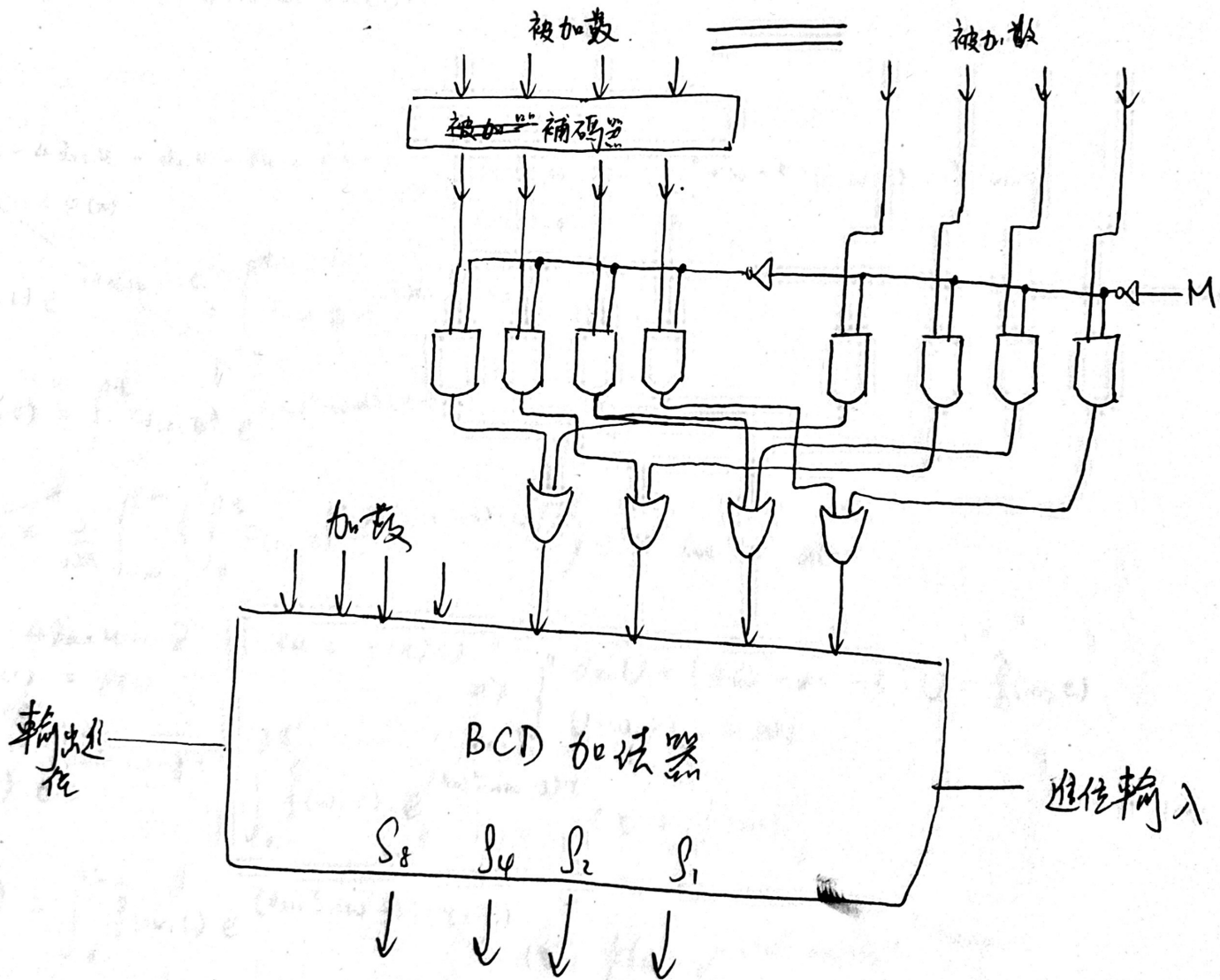
$$R_2 = S_1 S_2' + S_1' S_2 = S_1 \oplus S_2$$

$$R_1 = S_1$$

續 (1).



2. 當 $M=0$ 時, 是加法器; 當 $M=1$ 時, 是減法器



3.

$$t_{\text{REG1, setup}} = t_{\text{setup}} + t_{\text{buf}} = \cancel{0.7 \text{ ns}} \quad 0.7 \text{ ns}$$

$$t_{\text{REG2, hold}} = t_{\text{hold}} + t_{\text{buf}} = \cancel{0.2 \text{ ns}} \quad \cancel{0.4 \text{ ns}} \quad 0 \text{ ns}$$

$$t_{\text{REG3, pcg}} = t_{\text{pcg}} = 0.5 \text{ ns}$$

$$t_{\text{REG2, setup}} = t_{\text{setup}} - t_{\text{buf}} = 0.3 \text{ ns}$$

$$t_{\text{REG2, hold}} = t_{\text{hold}} + t_{\text{buf}} = 0.4 \text{ ns}$$

$$t_{\text{REG2, pcg}} = t_{\text{pcg}} + t_{\text{buf}} = 0.7 \text{ ns}$$

$$t_{\text{REG3, setup}} = t_{\text{setup}} = 0.5 \text{ ns}$$

$$t_{\text{REG3, hold}} = t_{\text{hold}} = 0.2 \text{ ns}$$

$$t_{\text{REG3, pcg}} = t_{\text{pcg}} + t_{\text{buf}} = 0.7 \text{ ns}$$